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The EPS Research Astro-RAG Platform: A Unified Open-Science Infrastructure for Cross-Epoch Astrophysical Kinematic Analysis, LLM-Assisted Research Workflows, and Educational Outreach

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ABSTRACT

We present the EPS Research Astro-RAG Platform (v1.0), an open-science research infrastructure providing four machine-readable, cross-epoch astrophysical corpora, 120 verified executable Jupyter notebook examples, a QuickStart reproducibility pathway, an educational High-School Exploration Track, and a roadmap for LLM-assisted retrieval-augmented generation (RAG) research workflows. The platform spans the full observable cosmic epoch from the local universe ($z = 0$) to the epoch approaching reionization ($z \sim 6$), providing the first unified kinematic dataset connecting HI 21cm rotation curves, Milky Way globular cluster dynamics, and high-redshift ALMA [CII] morpho-kinematics under a common schema and analysis framework. The four corpora contain 772 total objects: 438 HI-traced galaxies (Unified HI Rotation Curve Corpus v7.0), 129 dwarf/irregular galaxies (Dwarf/Irregular HI Corpus v1.0), 174 Milky Way globular clusters (GC Corpus v1.3.1), and 31 star-forming galaxies at $z = 4.26$–5.68 (High-z Kinematic Corpus Z1). The Unified HI corpus preserves the full fidelity of the Case Western Reserve University SPARC database (Lelli et al. 2016), including all photometric decomposition components ($V_{\rm gas}$, $V_{\rm disk}$, $V_{\rm bul}$) and surface brightness profiles, augmented with THINGS, LITTLE THINGS, and WALLABY DR2 data. The unifying scientific framework is the omega kinematic correction of Flynn & Cannaliato (2025), which reveals a robust sign reversal in the boundary-point angular velocity ω from positive values at $z = 0$ (SPARC mean $+7.06 \pm 3.26$ rad Gyr^{-1}) to negative values at $z \sim 5$ (Z1 median -13.05 rad Gyr^{-1}), motivating future RAMSES cosmological simulations. All corpora are released under CC BY 4.0 at Zenodo with permanent DOIs, and the platform is archived at https://github.com/eps-research/rag-corpus-series (DOI: 10.5281/zenodo.20398430).

Keywords: catalogs — methods: data analysis — galaxies: kinematics and dynamics — galaxies: dwarf — globular clusters: general — galaxies: high-redshift

1. INTRODUCTION

The intersection of large-scale astrophysical surveys and machine-learning infrastructure has created both an opportunity and a challenge for the research community. Surveys such as SPARC (Lelli et al. 2016), THINGS (Walter et al. 2008), LITTLE THINGS (Oh et al. 2015), WALLABY (Koribalski et al. 2020), and ALPINE (Le Fèvre et al. 2020) have produced rich kinematic datasets spanning orders of magnitude in galaxy mass, morphology, and cosmic epoch. However, these datasets exist in heterogeneous formats across dozens of source papers, making systematic cross-survey and cross-epoch analysis difficult for individual researchers.

The emergence of large language models (LLMs) and retrieval-augmented generation (RAG) pipelines as scientific tools adds a further requirement: datasets must be not only machine-readable but *self-describing* — structured such that an LLM can ingest, query, and reason over the data without external preprocessing. Standard FITS tables and survey-specific ASCII formats do not meet this requirement.

We present the EPS Research Astro-RAG Platform, a unified open-science infrastructure that addresses both challenges simultaneously. The platform provides four corpora in a common schema (JSON + flat CSV + RAG-ready JSONL + per-object ZIP), 120 verified ex-

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ecutable Jupyter notebooks spanning all four datasets, a reproducible QuickStart pathway, and a High-School Exploration Track designed to make astrophysical research accessible to students without prior domain knowledge.

The unifying scientific contribution across all four corpora is the omega kinematic correction (Flynn & Cannaliato 2025), which introduces a boundary-point angular velocity parameter ω derived from the innermost and outermost resolved rings of each rotation curve. Applied across 84 SPARC quality-1 galaxies, the correction yields a mean $\omega = +7.06 \pm 3.26$ rad Gyr^{-1} at $z = 0$. Extended to 8 confirmed rotators at $z \sim 4$–6 via ALMA [CII] 3DBarolo models (Jones et al. 2021), the same parameter yields a median $\omega = -13.05$ rad Gyr^{-1} — a sign reversal of ~ 20 rad Gyr^{-1} across ~ 9 Gyr of cosmic evolution.

This paper describes the platform architecture (Section 2), the four corpora (Section 3), the example library and educational track (Section 4), the reproducibility infrastructure (Section 5), and the planned LLM fine-tune team (Section 6). Data access and citation information are provided in Section 7.

2. PLATFORM ARCHITECTURE

The platform is organized into five functional silos, reflecting the natural progression from raw data to scientific analysis to publication to AI-assisted workflows:

1. **Silo 1 — $z = 0$ Data**: Three local-universe corpora (HI galaxies, dwarf/irregular galaxies, Milky Way globular clusters).
2. **Silo 2 — $z \sim 6$ Data**: The High-z Kinematic Corpus Z1, providing ALPINE [CII] morpho-kinematics at $z = 4.26$–5.68.
3. **Silo 3 — Example Library**: 120 verified Jupyter notebooks plus the High-School Exploration Track.
4. **Silo 4 — Papers & Preprints**: The EPS Research publication arc with DOIs and arXiv identifiers.
5. **Silo 5 — LLMs & Tools**: Planned fine-tuned model adapters and RAG utilities (Stage 2).

2.1. *Common Schema Design*

All four corpora share a common design philosophy. Each corpus is distributed in four formats:

- **JSON**: Hierarchical, per-object records with full metadata and per-measurement data arrays. Self-describing field names with units embedded.
- **Flat CSV**: One row per object, suitable for pandas, R, and spreadsheet analysis.
- **JSONL**: One JSON object per line, optimized for LLM RAG pipelines and streaming ingestion.
- **Per-object ZIP**: Individual JSON files for each object, enabling targeted retrieval without loading the full corpus.

Quality tiers are explicit in all corpora. For the HI and dwarf corpora, Tier 1 objects have full per-ring kinematic data; Tier 2 objects have integrated parameters only. For the Z1 corpus, Tier 1 objects are confirmed 3DBarolo rotators with per-ring $V_{\rm rot}$ and σ profiles; Tier 2 objects have morpho-kinematic classifications only.

3. THE FOUR CORPORA

3.1. *Unified HI Rotation Curve Corpus v7.0*

The Unified HI Rotation Curve Corpus v7.0 (Flynn 2026a) contains 438 spatially resolved rotation curves from four major HI surveys: SPARC (175 galaxies; Lelli et al. 2016), THINGS (34 galaxies; Walter et al. 2008), LITTLE THINGS (26 galaxies; Oh et al. 2015), and WALLABY DR2 (203 galaxies; Koribalski et al. 2020).

SPARC full-fidelity preservation. The SPARC component of this corpus preserves the complete photometric decomposition provided by the Case Western Reserve University SPARC database (Lelli et al. 2016). For each of the 175 SPARC galaxies, the corpus retains all per-ring components: the observed rotation velocity $V_{\rm obs}$ with error δV, the gas contribution $V_{\rm gas}$, the stellar disk contribution $V_{\rm disk}$, the bulge contribution $V_{\rm bul}$ (where applicable), and the 3.6 μm (Spitzer IRAC) surface brightness profiles $\Sigma_{\rm disk}$ and $\Sigma_{\rm bul}$. This full-fidelity preservation enables baryonic mass decomposition analyses and Tully–Fisher relation studies without requiring access to the original survey database. The SPARC data have been cross-validated against the published SPARC tables (Lelli et al. 2016, Table 1 and online appendix) at the individual ring level.

The corpus includes 84 SPARC galaxies with the original Lelli et al. (2016) Q=1 quality flag (the "golden sample" used in Flynn & Cannaliato 2025), identified by cross-referencing with the original SPARC Q=1 designations published in Lelli et al. (2016).

3.2. *Dwarf/Irregular HI Corpus v1.0*

The Dwarf/Irregular HI Corpus v1.0 (Flynn 2026c) contains 129 dwarf and irregular galaxies from LVHIS (Koribalski et al. 2019), VLA-ANGST (Ott et al. 2012), LITTLE THINGS (Oh et al. 2015), and WALLABY

DR2 (Koribalski et al. 2020). The corpus focuses on the low-mass, dark-matter-dominated end of the galaxy mass function, providing a complement to the SPARC-dominated HI corpus.

Of 129 total galaxies, 24 pass the "omega-ready" criterion (quality tier 1, at least 5 resolved rings, and kinematically regular morphology), yielding a dwarf corpus median $\omega = +9.94$ rad Gyr^{-1} — higher than the SPARC mean, consistent with the expectation that dark-matter-dominated systems with rising outer rotation curves produce larger positive omega values.

3.3. *Milky Way Globular Cluster Corpus v1.3.1*

The Milky Way Globular Cluster Corpus v1.3.1 (Flynn 2026b) contains 174 Milky Way globular clusters assembled from four primary sources: the Harris (1996, 2010 edition) photometric catalog (Harris 1996), Vasiliev & Baumgardt (2021) Gaia EDR3 proper motions (Vasiliev & Baumgardt 2021), HST structural parameters, and Baumgardt & Hilker (2018) N-body dynamical models (Baumgardt & Hilker 2018). All four source catalogs are cross-matched by cluster identifier and harmonized into a single per-cluster JSON record.

3.4. *High-z Kinematic Corpus Z1*

The High-z Kinematic Corpus Z1 (Flynn 2026d) contains 31 ALPINE (Le Fèvre et al. 2020) star-forming galaxies at $z = 4.26$–5.68, drawn from the Jones et al. (2021) morpho-kinematic classification catalog (Jones et al. 2021). Eight galaxies are confirmed Tier 1 rotators with per-ring 3DBarolo rotation curves; 23 additional galaxies have morpho-kinematic classifications only (Tier 2).

The corpus provides the high-redshift anchor of the platform, enabling the first direct cross-epoch omega comparison between local HI rotation curves and ALMA [CII] kinematics near the epoch of reionization.

Cross-epoch result. All 8 Tier 1 Z1 rotators show negative omega values (median -13.05 rad Gyr^{-1}) under the Flynn & Cannaliato (2025) kinematic correction, contrasting with positive values at $z = 0$ (SPARC mean $+7.06 \pm 3.26$ rad Gyr^{-1}; dwarf median $+9.94$ rad Gyr^{-1}). This sign reversal of ~ 20 rad Gyr^{-1} across ~ 9 Gyr of cosmic evolution is consistent with the structural transformation from compact, centrally concentrated high-z star-forming systems to extended rotating disks at $z = 0$. RAMSES cosmological simulations tracing this evolution from $z = 6$ initial conditions are planned for 2026–2027.

4. EXAMPLE LIBRARY AND EDUCATIONAL TRACK

4.1. *120 Verified Jupyter Notebooks*

The platform provides 120 executable Jupyter notebooks organized into five groups, all verified to execute without errors via automated `nbconvert` end-to-end testing:

- **SPARC/HI Examples** (25 notebooks): Rotation curve plotting, baryonic decomposition, omega correction, WALLABY Tier 2 analysis.
- **Dwarf/Irregular Examples** (25 notebooks): Omega-ready galaxy selection, DDO154/DDO161 cross-analysis, LVHIS and VLA-ANGST comparisons.
- **Globular Cluster Examples** (25 notebooks): Proper motion queries, N-body mass modeling, APOGEE chemistry cross-matching, multi-survey parameter comparison.
- **High-z Examples** (25 notebooks): [CII] rotation curve analysis, ALPINE population statistics, Wisnioski et al. (2015; W15) disk criteria evaluation, cross-corpus omega bridge.
- **High-School Exploration Track** (20 notebooks): Two-track curriculum (Track A: ages 12–14; Track B: ages 15–18) introducing galaxy kinematics, dark matter, and the omega correction without requiring prior astrophysics background.

All notebooks require only `numpy`, `matplotlib`, and `jupyter` as dependencies, with `scipy` for statistical notebooks. No proprietary software, telescope archive accounts, or institutional access is required.

4.2. *High-School Exploration Track*

The High-School Exploration Track is a novel feature that distinguishes the EPS Research platform from standard data releases. Track A (10 notebooks) is designed for students aged 12–14 with no prior astrophysics knowledge, introducing concepts such as galaxy morphology, rotation curves, and dark matter through guided data exploration. Track B (10 notebooks) targets ages 15–18 with a calculus-ready audience, introducing the omega correction formula, RMSE as a model evaluation metric, and cross-epoch kinematics.

Both tracks load directly from the same Zenodo corpus files used by the research notebooks, ensuring that students interact with real, published scientific data rather than simplified approximations. Cloud access via Google Colab (planned Stage 2) will remove the hardware barrier entirely, enabling in-classroom use without requiring students to install software.

5. REPRODUCIBILITY INFRASTRUCTURE

5.1. *QuickStart Pathway*

The platform provides a `QuickStart.ipynb` notebook at the repository root that implements a "golden path" through all four corpora. Starting from a fresh clone, a user can reproduce the core cross-epoch omega sign reversal result in under 10 minutes using only:

```
git clone https://github.com/eps-research/
          rag-corpus-series
cd rag-corpus-series
pip install -r requirements.txt
python download_corpora.py
jupyter lab
```

The `download_corpora.py` script fetches all four corpus JSON files directly from their Zenodo DOIs and places them in the correct directories for both the QuickStart notebook and all 120 example notebooks.

5.2. *Verification Protocol*

All 120 notebooks were verified using `jupyter nbconvert --execute` with a 120-second timeout per notebook. The full end-to-end retest across all five notebook groups confirmed a 120/120 pass rate. Notebooks are executed against the actual Zenodo corpus files to ensure that all data access patterns are valid and all field names are correct.

6. PLANNED LLM FINE-TUNE TEAM

Stage 2 of the platform (planned 2026) will add a four-model fine-tuned LLM team covering hardware configurations from laptop to research cluster:

The platform's JSONL schema is explicitly designed for LLM RAG pipelines. Stage 2 will provide a four-tier fine-tuned model team spanning research-cluster to laptop hardware, covering parameter scales from ∼72B (research-grade retrieval) to ∼7B (in-classroom, CPU-compatible inference). All models will be fine-tuned on the four EPS corpora using Unsloth QLoRA, with adapter weights published to HuggingFace at `eps-research/` upon completion. A vision-capable model tier (∼72B) will support extraction of kinematic values directly from ALMA plots and rotation curve figures into structured JSONL.

7. DATA ACCESS AND CITATION

All four corpora are released under CC BY 4.0 at Zenodo:

- Unified HI Corpus v7.0: 10.5281/zenodo.19563417
- Dwarf/Irregular Corpus v1.0: 10.5281/zenodo.20320362
- GC Corpus v1.3.1: 10.5281/zenodo.19907765
- High-z Corpus Z1: 10.5281/zenodo.20369285

The platform software is archived at Zenodo (DOI: 10.5281/zenodo.20398430) and maintained at https://github.com/eps-research/rag-corpus-series.

Authors using any EPS Research corpus are requested to cite the relevant Zenodo data descriptor and the Flynn & Cannaliato (2025) omega correction paper (Flynn & Cannaliato 2025) if the omega framework is used.

8. SUMMARY

We have presented the EPS Research Astro-RAG Platform v1.0, providing:

1. Four open, machine-readable corpora totaling 772 objects spanning $z = 0$ to $z \sim 6$, with permanent Zenodo DOIs.
2. Full-fidelity preservation of the SPARC photometric decomposition database, including all baryonic components.
3. 120 verified executable Jupyter notebooks with a 120/120 end-to-end pass rate, covering rotation curve analysis, globular cluster dynamics, high-redshift kinematics, and educational outreach.
4. A QuickStart reproducibility pathway enabling independent verification of the cross-epoch omega sign reversal in under 10 minutes.
5. A High-School Exploration Track making real published astrophysical data accessible to students aged 12–18 without prior domain knowledge.
6. A documented roadmap for Stage 2 LLM fine-tune deployment on hardware ranging from free Google Colab to research-grade dual-GPU clusters.

The cross-epoch omega sign reversal — from $+7.06 \pm 3.26$ rad Gyr^{-1} at $z = 0$ to -13.05 rad Gyr^{-1} at $z \sim 5$ — is the unifying scientific result motivating the platform's design and its planned RAMSES cosmological simulation program.

DECLARATION OF GENERATIVE AI USE

In accordance with standard practice for AI-assisted research, the author discloses the following. Four large language models were used during the creation and validation of this corpus and manuscript: Google Gemini

Pro, Anthropic Claude (Sonnet and Opus), Microsoft Copilot Pro, and Gemma 4 31B Dense (self-hosted). Because the corpus is explicitly designed for LLM-based RAG pipelines, these models served as both development tools and validation instruments. All scientific content, data analysis, corpus construction, and conclusions are the sole responsibility of the author.

The author thanks J. Cannaliato for collaboration on the omega correction framework. This research was conducted independently at EPS Research without institutional affiliation or external funding. All data products are released openly under CC BY 4.0.

Facility: WSRT, VLA, ALMA, HST, Gaia

Software: numpy (Harris et al. 2020), matplotlib (Hunter 2007), astropy (Astropy Collaboration et al. 2022), 3DBarolo (Di Teodoro & Fraternali 2015), jupyter (Kluyver et al. 2016)

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